

def throughput-and-latency(): (no gradient/optimize scores, since inference is memory-bound no training is needed)

Let's compute the theoretical maximum latency and throughput of a single request.

Initiate latency and throughput.

for Llama 2 3B on an H100

⇒ The memory, throughput, latency depends on shape of Transformer

$$\begin{cases} \text{memory} = B * \text{kv-cache-size} + \text{param-size} \\ \text{latency} = \text{memory} / \text{memory-bandwidth} = \text{determined by memory IO} \\ \text{throughput} = B / \text{latency} = \text{inverse of latency} \end{cases}$$

(generating B tokens in parallel)

if we use batch size of 1:

$$\text{memory} = 26,869, \dots, 000 \text{ KV cache}$$

$$\text{latency} = 0.00802 \text{ seconds/Token}$$

$$\text{throughput} = 124.67 \text{ tokens/s}$$

use batch size of 64: memory = 76,717, 990, 400, more kv cache

$$\text{latency} = 0.0238 = \text{higher } \uparrow$$

$$\text{throughput} = 2689.48 = \text{higher } \uparrow$$

Trade off: latency (VS) throughput:

- 1: smaller batch size yields better latency but worse throughput
- 2: larger batch size yields better throughput but worse latency

Solutions:

- launch M copies of Model, latency is the same, throughput increases by M
- shard the Model and the KV cache

Notes:

- TTFT is important for prefill
- use smaller batch sizes during prefill for faster TTFT
- use larger batch sizes during generation to improve throughput

Taking shortcuts (lossy)

def reduce-KV-cache-size():

reduce KV-cache size without sacrificing much accuracy

traditional: multi-head

(Llama 3)

Grouped-query

attention

multiple query

attention

values [] [] [] []

[] [] [] []

[]

keys [] [] [] []

share

KVs across heads

[] [] [] []

[] [] [] []

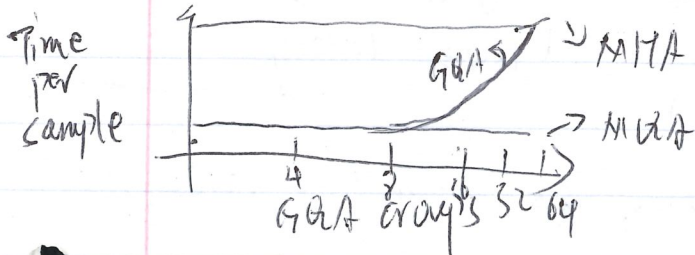
Queries [] [] [] []

(MHA)

(GQA)

(MHA)

latency / throughput improvements



e.g.,

Llama 3

similar accuracy

better throughput

⇒ Reduce the KV cache by a factor of N/K (worse latency)

e.g., (15)

MHA keys: 40

→

GQA keys: 8

memory: 79 → 33, xxx

latency & throughput: 2659 → 6416